

Shu Hu

PhD Student · Department of Microbiology · Fudan University

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EDUCATIONAL BACKGROUND

Fudan University

Sep 2025 – Jun 2029 (expected)

PhD student in Microbiology | Department of Microbiology, School of Life Sciences

The University of Edinburgh

Sep 2021 – Nov 2022

MSc in Bioinformatics, with Merit

Sichuan Agricultural University

Sep 2016 – Jun 2020

BSc in Biological Science | GPA: 3.8/5.0; average mark: 88/100.

English: CET-6 535; IELTS 6.0.

RESEARCH FOCUS: Foundation models and deep learning for biological sequences, with applications in viral evolution, antibody–virus interactions and genomic epidemiology.

PUBLICATIONS

- [1] **Hu S**, Ye D, Liao M, Xu T, Fan T, Dong N, Cui J. Paired Env-antibody representation learning predicts HIV-1 neutralization and traces antibody escape across viral evolution. *Cell Reports Methods*. 2026. **Under revision**
- [2] Ye D, Hu F, Hu H, **Hu S**, Tan Y, Ouyang W, Li SZ, Cui J, Dong N. ViroBench: Benchmarking Nucleotide Foundation Models on Viral Genomics Tasks. *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2*. 2026; 12679–12690. doi: [10.1145/3770855.3819057](https://doi.org/10.1145/3770855.3819057)
- [3] Schwab LSU, Xie R, Reilly E, Aban M, **Hu S**, Spirason N, Deng YM, et al. Epistasis facilitates the long-term antigenic evolution of the influenza B virus hemagglutinin. *bioRxiv*. 2026. doi: [10.64898/2026.07.22.739994](https://doi.org/10.64898/2026.07.22.739994)
- [4] Xie R, Zhukova A, Peña PG, Iglesias G, **Hu S**, Wang J, Tsang TK, Dhanasekaran V, Kraemer MUG, Pybus OG, Gascuel O. Scalable deep-learning-based inference of time-varying transmission dynamics from outbreak phylogenies. *medRxiv*. 2026. doi: [10.64898/2026.05.07.26352673](https://doi.org/10.64898/2026.05.07.26352673)
- [5] Siegers JY, Xie R, Edwards KM, Byrne AMP, **Hu S**, et al. Resurgence of Zoonotic Highly Pathogenic Avian Influenza A(H5N1) Virus in Cambodia. *New England Journal of Medicine*. 2025; 393(16):1650–1652. doi: [10.1056/NEJMc2504302](https://doi.org/10.1056/NEJMc2504302)
- [6] **Hu S**, Cui J. Cattle H5N1 outbreak in US driven by cow’s “cross-nursing” behaviour: elucidating a novel transmission mechanism. *Emerging Microbes & Infections*. 2025; 14(1):2547736. doi: [10.1080/22221751.2025.2547736](https://doi.org/10.1080/22221751.2025.2547736)
- [7] Yang K, Nizami S, **Hu S**, Zou L, Deng H, Xie J, Guo Q, Edwards KM, Dhanasekaran V, Yen HL, Wu J. Genetic diversity of highly pathogenic avian influenza H5N6 and H5N8 viruses in poultry markets in Guangdong, China, 2020–2022. *Journal of Virology*. 2025; 99(1):e01145–24. doi: [10.1128/jvi.01145-24](https://doi.org/10.1128/jvi.01145-24)
- [8] Xie R, Adam DC, **Hu S**, Cowling BJ, Gascuel O, Zhukova A, Dhanasekaran V. Integrating Contact Tracing Data to Enhance Outbreak Phylodynamic Inference: A Deep Learning Approach. *Molecular Biology and Evolution*. 2024; 41(11):msae232. doi: [10.1093/molbev/msae232](https://doi.org/10.1093/molbev/msae232)

RESEARCH AND PROFESSIONAL EXPERIENCE

Fudan University

Sep 2025 – present

PhD Research

- Co-developed UniCAV for HIV-1 neutralization prediction using protein language models; curated data, trained and benchmarked models, and drafted the manuscript (Cell Reports Methods, under revision).
- Contributed to ViroBench, a benchmark of nucleotide foundation models for viral genomics (KDD 2026).

The University of Hong Kong

Jun 2023 – Jun 2024

Research Assistant I

- Contributed to deep-learning phylodynamics research combining contact-tracing and phylogenetic data; co-authored the resulting Molecular Biology and Evolution paper (2024).
- Analysed influenza and RSV genomic data; contributed to avian-influenza genomic epidemiology studies in Cambodia and southern China (NEJM and Journal of Virology, 2025).

Shanghai Rightongene Technology Co., Ltd.

2024 – May 2025; Sep 2022 – May 2023

Bioinformatics Engineer

- Developed gene-fusion analysis methods for BCL6, BCL2 and MYC in DLBCL sequencing workflows, addressing discrepancies between NGS and FISH results.
- Established low-depth WGS copy-number analysis workflows using QDNAseq, Control-FREEC and ichorCNA.
- Built configurable single-cell RNA-seq workflows in Shell and R; collaborated with IT colleagues on automated platform deployment.
- Developed RNA-seq expression-based models for leukemia molecular subtyping.

The University of Edinburgh

Feb 2022 – Sep 2022

MSc Research

- Developed reusable R Shiny modules and scripts for interactive exploration of RNA-seq data.

Sichuan WoWente Biotechnology Co., Ltd.

May 2020 – May 2021

Reagent Developer

- Contributed to sex-hormone immunoassay reagent development and an inhibin A (INH-A) project.

TECHNICAL SKILLS

Foundation models: Protein language models (ESM-C, AntiBERTa2); nucleotide model benchmarking; biological sequence representation learning.

Deep learning: PyTorch, Hugging Face Transformers, scikit-learn; model development, training, evaluation and interpretation.

Programming: Python, R (Shiny), Shell, Linux, Docker, SQL, Git.

Bioinformatics: Phylogenetics and phylodynamics (BEAST, Nextstrain); RNA-seq, single-cell RNA-seq, WGS copy-number analysis and gene-fusion detection.

RESEARCH SOFTWARE

github.com/hushuzuiniu

UniCAV · Single-cell RNA-seq workflows · RNA-seq molecular subtyping · R Shiny tools